Benchmarking Continuous Dynamic Optimization: Survey and Generalized Test Suite

Danial Yazdani, *Member, IEEE*, Mohammad Nabi Omidvar, *Member, IEEE*, Ran Cheng, *Member, IEEE*, Jürgen Branke, *Member, IEEE*, Trung Thanh Nguyen, and Xin Yao, *Fellow, IEEE*

***Abstract*—Dynamic changes are an important and inescapable aspect of many real-world optimization problems. Designing algorithms to find and track desirable solutions while facing challenges of dynamic optimization problems is an active research topic in the field of swarm and evolutionary computation. To evaluate and compare the performance of algorithms, it is imperative to use a suitable benchmark that generates problem instances with different controllable characteristics. In this article, we give a comprehensive review of existing benchmarks and investigate their shortcomings in capturing different problem features. We then propose a highly configurable benchmark suite, the generalized moving peaks benchmark, capable of generating problem instances whose components have a variety of properties, such as different levels of ill-conditioning, variable interactions, shape, and complexity. Moreover, components generated by the proposed benchmark can be highly dynamic with respect to the gradients, heights, optimum locations, condition numbers, shapes, complexities, and variable interactions. Finally, several well-known optimizers and dynamic optimization algorithms are chosen to solve generated problems by the proposed benchmark. The experimental results show the poor performance of the existing methods in facing new challenges posed by the addition of new properties.**

***Index Terms*—Dynamic environments, dynamic optimization problems (DOPs), evolutionary dynamic optimization, moving peaks benchmark (MPB), survey, tracking moving optimum (TMO).**

Manuscript received September 6, 2019; revised January 27, 2020; accepted July 9, 2020. Date of publication August 14, 2020; date of current version May 19, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 61903178 and Grant 61906081; in part by the Program for Guangdong Introducing Innovative and Enterpreneurial Teams under Grant 2017ZT07X386; in part by the Shenzhen Peacock Plan under Grant KQTD2016112514355531; in part by the Guangdong Provincial Key Laboratory under Grant 2020B121201001; and in part by the Program for University Key Laboratory of Guangdong Province under Grant 2017KSYS008. This article was recommended by Associate Editor G. G. Yen. *(Corresponding author: Ran Cheng.)*

Danial Yazdani and Ran Cheng are with the Guangdong Provincial Key Laboratory of Brain-Inspired Intelligent Computation, Department of Computer Science and Engineering, Southern University of Science and Technology, Shenzhen 518055, China (e-mail: danial.yazdani@gmail.com; chengr@sustc.edu.cn).

Mohammad Nabi Omidvar is with the School of Computing, University of Leeds, Leeds LS2 9JT, U.K., and also with the Business School, Leeds University, Leeds LS2 9JT, U.K. (e-mail: m.n.omidvar@leeds.ac.uk).

Jürgen Branke is with the Operational Research and Management Sciences Group, Warwick Business School, University of Warwick, Coventry CV4 7AL, U.K. (e-mail: juergen.branke@wbs.ac.uk).

Trung Thanh Nguyen is with the Department of Maritime and Mechanical Engineering, Liverpool John Moores University, Liverpool L3 3AF, U.K. (e-mail: t.t.nguyen@ljmu.ac.uk).

Xin Yao is with the Guangdong Provincial Key Laboratory of Brain-Inspired Intelligent Computation, Department of Computer Science and Engineering, Southern University of Science and Technology, Shenzhen 518055, China, and also with CERCIA, School of Computer Science, University of Birmingham, Birmingham B15 2TT, U.K. (e-mail: xiny@sustc.edu.cn).

This article has supplementary material provided by the authors and color versions of one or more figures available at https://doi.org/10.1109/TCYB.2020.3011828.

Digital Object Identifier 10.1109/TCYB.2020.3011828

I. Introduction

SEARCH spaces of many real-world optimization problems are dynamic in terms of the objective function, the number of variables, and/or constraints [1]. Optimization problems that change over time and need to be solved online by an optimization method are referred to as dynamic optimization problems (DOPs) [2]. To solve DOPs, algorithms not only need to find desirable solutions but also to react to the environmental changes in order to quickly find a new solution when the previous one becomes suboptimal [3].

To comprehensively evaluate the effectiveness of algorithms designed for DOPs, a suitable benchmark generator is crucial. A DOP benchmark generator should have the following major features [1], [2].

1) *Easy to Implement and Analyze:* Researchers should have access to information about the location of the global optimum and its fitness value, and characteristics of components that form the landscape, such as their variable interaction structure, shape, and change intensities. Although this information should not be used for designing/tuning algorithms and the problem must remain a black box for algorithms, it allows researchers to analyze and measure the behavior and performance of the algorithms. In addition, a good benchmark should not be excessively difficult to implement.
2) *Flexibility:* The benchmark generator should be highly configurable with respect to different aspects, such as the number of components, shape of components, dimension, change frequency, and change severity. In addition, the configuration of any feature should be independent of other features. This independence allows the researchers to investigate the effect of every single property on the behavior of algorithms. Moreover, a good benchmark should be capable of generating problem instances with varying degrees of complexity. Many real-world problems are very hard and complex [1], [2], [4].

2168-2267 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on October 19,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

3) *Variety:* The benchmark generator should be capable of generating problems instances with a variety of characteristics, such as modularity (fully separable, fully nonseparable, and partially separable), components with different condition numbers (ill-conditioning), different intensity of local and global modality (unimodal to highly multimodal), heterogeneity, balanced to highly imbalanced subfunctions, low to high dimensional (large scale), different levels of irregularities (smooth to highly irregular), and symmetric components to highly asymmetric ones.

Several DOP benchmarks have been proposed in the literature for different DOP domains [2], [5], such as combinatorial [6], [7]; continuous [4], [8]; multiobjective [9]–[11]; and constrained DOPs [12]–[14]. This article focuses on dynamic continuous unconstrained single-objective optimization problems. For brevity, we use the term DOP to refer to the considered DOPs in this article. Commonly used and well-known benchmark generators in this domain use the idea of having several components that form the landscape. In most existing DOP benchmarks, the width, height, and location of these components change over time [5]. One of the benchmark generators that is designed based on this idea is the moving peaks benchmark (MPB) [8], which is the most widely used synthetic problem in the DOP field [1], [3], [5]. Each component in MPB is formed by usually a simple peak whose basin of attraction is determined using a $\max(\cdot)$ function. MPB and its variations have been used in different classes of DOPs, such as tracking moving optimum (TMO) in which the algorithms search for the best solution in each environment [2]; robust optimization over time (ROOT) in which the algorithms search for solutions whose quality remain acceptable after environmental changes [15], [16]; multiobjective DOPs in which there are multiple conflicting objectives [17]; large-scale DOPs, where algorithms face the scalability issues due to high dimensionality of the problem [18], [19]; and DOPs with dynamic constraints in which there are several moving feasible regions in the landscape [13].

Despite the popularity of the standard MPB, landscapes generated by MPB consist of components that are smooth, symmetric, unimodal, separable [18], and easy to optimize, which may not be the case in many real-world problems. Note that MPB is capable of using basic functions, such as Rastrigin, Griewank, Ackley, and Rosenbrock as its components instead of the simple peaks. However, practically it is very challenging to use the aforementioned basic functions due to their different characteristics, such as shape, gradient, the ratio between the fitness value drop to the distance to the optimum, and standard search range. In fact, many basic functions cannot be used as the components of MPB, and for each of the ones that can be utilized, all of the MPB parameters need to be specifically tuned. On the other hand, using different basic functions in the MPB to have components with different characteristics is more challenging and almost impossible in many cases.

In this article, we propose a *generalized* MPB (GMPB) which is capable of generating problems with a variety of characteristics that can range from fully nonseparable to fully separable structure, from homogeneous to highly heterogeneous subfunctions, from balanced to highly imbalanced subfunctions, from unimodal to highly multimodal components, from symmetric to highly asymmetric components, and from smooth to highly irregular components. Therefore, GMPB is a benchmark generator with fully controllable features that helps researchers analyze DOP algorithms and investigate their efficiencies in facing a variety of different problem characteristics. Although the aforementioned problem characteristics have been vastly investigated in other fields of evolutionary computations, such as static optimization problems [20]–[22] and large-scale optimization problems [23], [24], little related work has been dedicated to the DOP literature.

The major contributions of this article can be summarized as follows.

1) A comprehensive survey on continuous unconstrained single-objective DOP benchmarks that investigates their baselines, dynamics, characteristics, and behaviors. This survey helps the reader to understand the shortcomings of the existing benchmarks.
2) A new benchmark with the capability of generating a variety of problem characteristics that have not been considered in the existing DOP benchmarks. All the characteristics are mathematically formulated into a baseline function that can be easily configured and controlled through its parameters. GMPB is capable of generating problems with any desired combination of characteristics that can also change over time.
3) A detailed analysis of the behavior of several well-known algorithms on different problem characteristics.

Background information, including the definitions of DOPs, variable interaction, modularity, imbalance, heterogeneity, and ill-conditioning, are provided in Section S-I of the supplementary document. The organization of the remainder of this article is as follows. Section II provides a survey of continuous unconstrained single-objective DOP benchmarks. Section III proposes the GMPB. Section S-II in the supplementary material of the supplementary material conducts a comprehensive empirical analysis of the GMPB. Finally, Section IV concludes this article and outlines possible future directions.

II. Survey on DOP Benchmarks

In this section, we review the commonly used and well-known DOP benchmarks, which are continuous, single objective, and unconstrained. In this article, DOP benchmarks are categorized as follows.

1) DOP benchmarks whose landscapes are formed by a single component.
2) DOP benchmarks whose landscapes are made by jointing several components. The benchmarks belonging to this category handle multiple components in two major ways:
   a) using the $\max(\cdot)$ function to control the basin of attraction among multiple components;
   b) partitioning the search space into hypercube-based components.

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on October 19,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

A. DOP Benchmarks Whose Landscape Is Formed by Single Component

The AB benchmark [25] is the first in this category, which uses two 2-D landscapes called *A* and *B*, each with different characteristics. The *A* landscape consists of 14 sinusoidally shaped peaks, and *B* consists of 20 peaks, whose shapes are determined by a sine function, and triangular and Gaussian probability distributions. Three types of environmental changes are defined as follows: 1) linear translation of peaks in *A*; 2) changing the location of the maximum peak randomly while the rest of the search space remains unchanged; and 3) switching between landscapes *A* and *B*.

Moving parabola (MP) [26] applies linear, circular, and random dynamics to a simple 3-D sphere function to shift its location. Given a severity parameter $\tilde{\phi}$, the aforementioned three dynamics are formulated as follows:

$$\phi_i^{(t+1)} = \phi_i^{(t)} + \tilde{\phi} \tag{1}$$

$$\phi_i^{(t+1)} = \begin{cases} \phi_i^{(t)} + \tilde{\phi} \sin \frac{2\pi t}{25} & i = 1, 3 \\ \phi_i^{(t)} + \tilde{\phi} \cos \frac{2\pi t}{25} & i = 2 \end{cases} \tag{2}$$

$$\phi_i^{(t+1)} = \phi_i^{(t)} + \tilde{\phi}\mathcal{N}(0, 1) \tag{3}$$

where $\phi_i^{(t)}$ is the offset of the ith dimension in the t-th environment for $i \in \{1, 2, 3\}$, $\mathcal{N}(0, 1)$ is a random number drawn from a Gaussian distribution with mean 0 and variance 1, and $\tilde{\phi}$ is the change severity.

Dynamic rotation (DR) [27] combines the landscape with a visibility mask, which only allows the fitness of certain regions of the landscape to be evaluated. The visibility mask forces the remaining regions to return a predefined undesirable fitness value. DR uses five types of dynamics: 1) rotating the objective function around the optimum where there is no visibility mask; 2) rotating the visibility mask around the optimum where the objective function is stationary; 3) rotating the objective function around its optimum while the visibility mask is static; 4) rotating both the objective function and the visibility mask around the optimum with the same rotation angle; and 5) rotating both the objective function and the visibility mask around the optimum with different rotation angles.

The composition dynamic benchmark generator (CDBG) [28] is extended from the static composition functions in [22] and [21]. Since the landscape of CDBG is made by composing several single-component subfunctions, we cannot categorize it as a benchmark with joining components. Moreover, since it uses some standard static basic functions, such as Ackley and Rastrigin in its structure, the number of peaks cannot be controlled by the user. CDBG uses the environmental dynamics of the generalized dynamic benchmark generator (GDBG) [28] for shifting each basic function, which will be discussed later in this section.

A general framework for inducing artificial changes (IACs) in optimization problems is proposed in [29]. The utilized landscapes in IAC contain some basic functions, such as Rosenbrock, Griewank, Ackley, and Rastrigin. IAC contains four groups of dynamics that can be applied to the landscapes to generate environmental changes. These dynamics include: 1) rotation; 2) shift; 3) decision variables permutation; and 4) duplication-based dynamics where a predefined number of decision variables are randomly copied to other positions. The aforementioned dynamics can be applied to the entire landscape, or to some regions of the search space. Consequently, IAC can generate both local or global environmental changes.

B. DOP Benchmarks Whose Landscape Is Formed by Joining Several Components

This group of DOP benchmarks is further categorized into the ones whose landscape is divided using the $\max(\cdot)$ function or partitioning by hypercubes.

1) DOP Benchmarks Based on the max(·) Function: This group of DOP benchmarks uses a $\max(\cdot)$ function to define the basin of attraction of components, which are usually single peaks. Branke's MPB [8] is the most popular benchmark suite in DOPs [1], [5]. MPB generates a landscape containing several components in which each component contains a peak whose height, width, and location change every time the environment changes. MPB is flexible and can be used to generate scalable functions with a configurable number of components. Each component has the potential to become the global optimum after an environmental change. In the first version of MPB [8], the baseline function was defined as follows:

$$f^{(t)}(\mathbf{x}) = \max_{i \in \{1,\ldots,m\}} \frac{h_i^{(t)}}{1 + w_i^{(t)} \sum_{j=1}^{d} \left(\mathbf{x}_j - \mathbf{c}_{i,j}^{(t)}\right)^2} \tag{4}$$

where m is the number of components (peaks), $\mathbf{x}$ is a solution in the d-dimensional problem space, and $h_i^{(t)}$, $w_i^{(t)}$, and $\mathbf{c}_i^{(t)}$ are the height, width, and the center of the ith peak in the t-th environment, respectively. Later, a new scenario of MPB was proposed (Scenario 2), which replaces the *peak* function in (4) with the *cone* function [30] (as in the DF1 benchmark generator [31], [32]). Scenario 2 has become the standard configuration of MPB in many studies [33]–[41].

The Scenario 2 baseline function of MPB is as follows:

$$f^{(t)}(\mathbf{x}) = \max_{i \in \{1,\ldots,m\}} \{h_i^{(t)} - w_i^{(t)} \|\mathbf{x} - \mathbf{c}_i^{(t)}\|\}. \tag{5}$$

Fig. 1(a) and (b) illustrates MPB landscapes generated by (4) and (5), respectively. In MPB, the height, width, and center of all components change from one environment to the next based on the following update rules:

$$h_i^{(t+1)} = h_i^{(t)} + \tilde{h}\mathcal{N}(0, 1) \tag{6}$$

$$w_i^{(t+1)} = w_i^{(t)} + \tilde{w}\mathcal{N}(0, 1) \tag{7}$$

$$\mathbf{c}_i^{(t+1)} = \mathbf{c}_i^{(t)} + \mathbf{v}_i^{(t+1)} \tag{8}$$

$$\mathbf{v}_i^{(t+1)} = \tilde{s} \cdot \frac{(1 - \lambda) \cdot \mathbf{u} + \lambda \cdot \mathbf{v}_i^{(t)}}{\left\|(1 - \lambda) \cdot \mathbf{u} + \lambda \cdot \mathbf{v}_i^{(t)}\right\|} \tag{9}$$

where $\mathcal{N}(0, 1)$ is a random number drawn from a Gaussian distribution with mean 0 and variance 1, $\tilde{h}$ is the height severity, $\tilde{w}$ is the width severity, $\tilde{s}$ is the shift severity, $\mathbf{u}$ is a vector of uniformly distributed numbers in range $[-0.5, 0.5]$, and $\lambda \in (0, 1)$ is the correlation coefficient. $\lambda = 0$ implies that the peak relocations are uncorrelated whereas they are fully

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on October 19,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

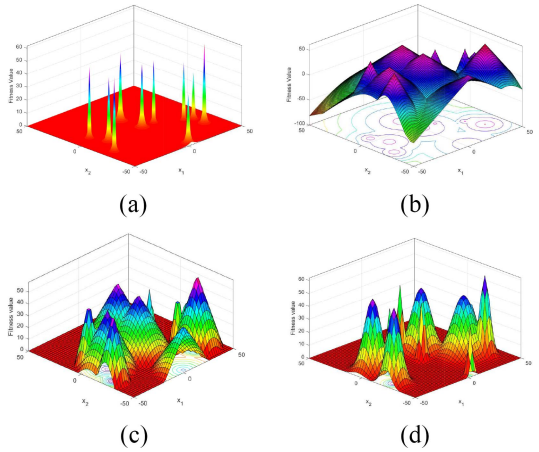


Fig. 1. Landscapes with ten components generated by different baselines. All shared parameter settings, such as component centers' locations, heights, and widths, are equal. In Fig. 1(c), β is set to zero. Note that in contrary to the flat regions in Fig. 1(c) whose gradient is zero, the gradients in the flat looking regions in Fig. 1(a) and (d) have positive nonzero values. (a) Landscape generated by (4). (b) Landscape generated by (5). (c) Landscape generated by (10). (d) Landscape generated by (12).

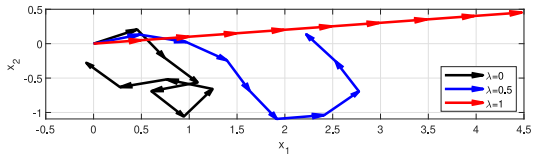


Fig. 2. Movement of the peak center through a 2-D subspace over ten environmental changes where $\tilde{s} = 1$. The initial position of peak is [0, 0], and the same random seed is used for each sequence of movements.

correlated for $\lambda = 1$. The step size of peak movements is constant and equal to $\tilde{s}$. Fig. 2 shows ten relocations of a peak center using different values of λ, where $\tilde{s} = 1$. According to (6) and (7), the default environmental dynamic for a peak's height and width in MPB is *random change* [26] based on (3). At each environmental change, all components' parameters change using (6)–(8). Consequently, the entire search space changes after each environmental change. Considering this characteristic, an environmental change can be detected by re-evaluating any solution across the search space by comparing its current and previous fitness values. If the values are different, environmental change can be detected.

In [42], a control parameter is added to MPB to determine the percentage of changing components at each environmental change. As a result, a predefined number of randomly chosen components will change using (6)–(8), while the remaining ones will remain unchanged. Therefore, change detection becomes more challenging for some methods as only some parts of the environment change [40]. Furthermore, in order to increase the size of the regions that remain unchanged after environmental changes, a threshold β is used

$$f^{(t)}(\mathbf{x}) = \max\left\{\beta, \max_{i\in\{1,\ldots,m\}}\left\{h_i^{(t)} - w_i^{(t)}\|\mathbf{x} - \mathbf{c}_i^{(t)}\|\right\}\right\} \quad (10)$$

where β determines the minimum fitness value of the problem and creates flat regions. The fitness value of any solution in these flat regions is equal to β. These flat regions make the problem more challenging since the change detection has become harder. Fig. 1(c) shows a generated landscape using (10).

In the field of ROOT [15], the MPB with the baseline (5) is used in many studies [16], [43]–[47]. However, the values of $\tilde{h}$, $\tilde{w}$, and $\tilde{s}$ are different for each component. The reason for having different height, width, and shift severities for each component is to create components with different levels of robustness. Equations (3), (14), and (18) are used as dynamics in [47] while default MPB dynamics, that is, (3), are used in [16] and [43]–[46].

The MPB has one global optimum in each environment. However, most multipopulation methods [2] try to cover and track multiple optima. Note that although multiple optima are covered by many DOP algorithms [2], most DOPs only have single global optimum, despite that there may also exist local optima. This is different from multimodal optimization where there exist multiple global optima inside the same fitness landscape [48], [49]. In many DOP algorithms, especially multipopulation methods, covering multiple optima is done in order to increase and maintain overall diversity, and accelerate the process of finding new global optimum after environmental changes. In [50], a multimodal version of MPB with multiple global optima is proposed. In this benchmark, a predefined number of components are global optima whose heights are constant over time, and only their locations and widths are time varying. There are other local optima whose heights change over time. However, the upper bound of the height value of the local optima is less than that of the global optima.

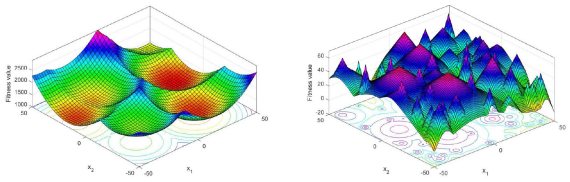
Pendulum MPB (PMPB) [51] is a modified version of MPB in which the environments reappear periodically over time. PMPB works based on a pendulum length parameter *pl* and a direction parameter *dir*. First, using the baseline (5) and the dynamics from (6)–(8), *pl* environments $[f^{(1)}, f^{(2)}, \ldots, f^{(pl)}]$ are constructed. Then, the environmental parameters of the constructed environments are stored in an archive to be used for future reappearances. *dir* shows the direction of fetching the environmental parameters from the archive, to retrieve the next environment. The value of *dir* changes every *pl* environmental changes by switching between *right* and *left*. The environments are retrieved from the archive based on their order and the value of *dir*. An example of the pendulum-based environmental changes is

$$\left[f^{(1)}, f^{(2)}, \ldots, f^{(pl-1)}, f^{(pl)}, f^{(pl-1)}, \ldots, f^{(2)}, f^{(1)}, f^{(2)}, \ldots\right]. \quad (11)$$

Similar to MPB, DF1 [31], [32] generates problem instances in which the width, height, and location of components change over time. The baseline function of DF1 is similar to (5); however, the dynamics are different and a logistic function is used to generate dynamics. Another benchmark, whose landscape consists of several components, is the Gaussian peaks benchmark (GPB) [52] which uses the following baseline:

$$f^{(t)}(\mathbf{x}) = \max_{i\in\{1,\ldots,m\}}\left\{h_i^{(t)} \exp\left(-\frac{\|\mathbf{x} - \mathbf{c}_i^{(t)}\|^2}{2\left(w_i^{(t)}\right)^2}\right)\right\}. \quad (12)$$

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on October 19,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 3. Two examples in which some components are covered by larger ones. (a) Landscape generated by (13) with $m = 10$ and $m' = 6$. (b) Landscape generated by (5) with $m = 100$ and $m' = 68$.

An example of a generated 2-D landscape by this benchmark is shown in Fig. 1(d). In GPB, the locations of peaks change in random directions, and the step sizes are uniformly distributed over an interval controlled by two levels of severity called abrupt and gradual.

In contrast to the aforementioned landscapes, Blackwell [53] proposed a minimization benchmark generator denoted as moving valleys benchmark (MVB). The generated landscapes by MVB are constructed by several spheric components, whose basins of attractions are determined by a min(·) function. MVB is constructed using the following baseline function:

$$f^{(t)}(\mathbf{x}) = \min_{i\in\{1,\ldots,m\}} \left\{ \sum_{j=1}^{d} \left(\mathbf{x}_j - \mathbf{c}_{i,j}^{(t)}\right)^2 + \left(h_i^{(t)}\right)^2 \right\}. \quad (13)$$

As can be seen in (13), MVB does not use width parameter w which leads to construct components with identical gradients. Fig. 3(a) illustrates a 2-D landscape generated by this benchmark.

One main property of all aforementioned benchmarks in this part is the dynamic number of visible components m' in the landscape. A component is visible if it is not covered by any other larger component. When components are wider, the possibility of decreasing m' increases. Among the investigated benchmarks, MVB has very broad components, such that it is probable that the smaller components become hidden under larger ones. For example, as can be seen in Fig. 3(a), the number of components m is set to ten while the number of visible components m' is smaller. Furthermore, the number of visible components drops by increasing m. In such circumstances, the density of components is increased such that larger components can cover several smaller close components. Fig. 3(b) shows a generated landscape by (5) where the number of components m is set to 100. However, almost a third of them are invisible. Changing the number of visible components is a challenging property for algorithms that try to cover and track multiple moving components. This is due to the fact that the algorithm will lose its track when a component becomes hidden under a larger component in an environmental change. In addition to the property of the varying number of visible component m', a modified version of MPB is proposed in [54] where the number of components m is changing over time.

The GDBG [28] contains six different types of dynamics, including small step, large step, random, chaotic, recurrent, and recurrent with noise which are formulated, respectively, as follows:

$$\Delta\phi = \gamma r\tilde{\phi}\|\phi\| \quad (14)$$

$$\Delta\phi = \tilde{\phi}\|\phi\|(\gamma \operatorname{sign}(r) + r(\gamma_{\max} - \gamma)) \quad (15)$$

$$\Delta\phi = \tilde{\phi}\mathcal{N}(0,1) \quad (16)$$

$$\phi^{(t+1)} = A\phi^{(t)}\frac{1-\phi^{(t)}}{\|\phi\|} \quad (17)$$

$$\phi^{(t+1)} = \phi_{\min} + \frac{\|\phi\|}{2}\left(\sin\left(\frac{2\pi}{p}t + \varphi\right) + 1\right) \quad (18)$$

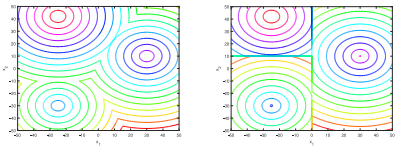
$$\phi^{(t+1)} = \phi_{\min} + \frac{\|\phi\|}{2}\left(\sin\left(\frac{2\pi}{p}t + \varphi\right) + 1\right) + \tilde{n}\mathcal{N}(0,1) \quad (19)$$

where $\Delta\phi$ is a deviation from the current control parameters, $\|\phi\|$ is the change range of ϕ, $\phi^{(t)}$ is the offset in the t-th environment, $\tilde{\phi} \in (0,1)$ is the change severity of ϕ, $\phi_{\min}$ is the minimum value of ϕ, $\tilde{n} \in (0,1)$ is the noise severity, $\gamma, \gamma_{\max} \in (0,1)$, and A are constant values, $r \in (-1,1)$ is a random number drawn from a uniform distribution, p is the period number, and φ is the initial phase. It is worth mentioning that the random change in (16) is the same as the dynamic that was used in MPB in (6) and (7), and in MP in (3). Moreover, the utilized logistic function for chaotic change in (17) is similar to the logistic function, which is used in DF1. In [55] and [56], other dynamics are added to GDBG, including change in the number of components and dimensions. Two further DOP benchmarks use the GDBG dynamics, including CDBG which was described in Section II-A, and rotation DBG (RDBG) [28]. RDBG is constructed based on the baseline of MPB in (4). The height and width of components change using a dynamic from GDBG. Furthermore, the location of components changes using rotation matrices introduced in DR [27]. Since the relocations of components by the rotation procedure are not controllable in RDBG, it is not capable of constructing cyclic environments. In [57], the procedure of rotation in RDGB is modified in order to control the components' relocation steps. This benchmark is capable of constructing cyclic environments with a predefined cycle length.

Although most DOP benchmarks are scalable, they cannot produce modular problems whose components contain multiple moving optima. To address this shortcoming, some studies used composition methods to create modular problems. High-dimensional MPB (HDMPB) [19] is the first modular DOP benchmark with multiple component subfunctions, which is built by the summation of several MPBs to create large-scale DOPs. CMPB [3], [18] is another benchmark that composes several MPBs to create modular problem instances. Unlike HDMPB, CMPB can generate partially separable problems that could contain fully separable dimensions. In addition, CMPB is capable of generating modular problems that are imbalanced and heterogeneous.

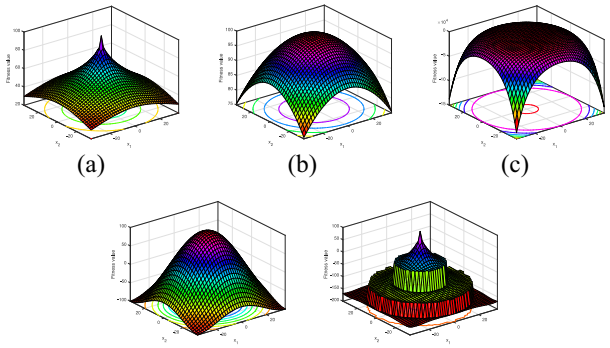
2) DOP Benchmarks Based on Hypercube Partitioning: The first DOP benchmark of this category is dividing search space (DSS) [58] that partitions the search space into subspaces. Each subspace contains a simple peak function whose summit is at the center of the subspace. For dividing the d-dimensional search space, each dimension is equally divided into k segments resulting in k^d subspaces.

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on October 19,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 4. Two landscapes with three peaks (components) whose basin of attraction is determined based on their hypercube, or max(·) function. The peak function is $s_1 = h - \|\mathbf{x} - \mathbf{c}\|$ from [4] which is a *cone* function from (5) with $w = 1$. (a) max(·)-function-based basin of attraction. (b) Hypercube-based basin of attraction.



(a) (b) (c) (d) (e)

Fig. 5. Five types of components that are used in FPs [4].

Another DOP benchmark in which the idea of space partitioning by hypercubes is free peaks (FPs) [4]. FPs is a complicated benchmark generator whose search space is divided into components using a k-d tree [59]. In each component, there is only one component that can move inside the hypercube. In FPs, the basin of attraction of components is determined by the hypercubes, which is different from other benchmarks, such as MPB, DF1, and GPB. An illustrative example is provided in Fig. 4 to show the difference between the two types of the basin of attractions. As can be seen in Fig. 4(b), the landscape generated by hypercube partitioning has sharp ridges around components. The shape of each component in FPs is chosen from eight different *unimodal* basic functions. From these eight basic functions, three of them are similar to the components (with $w = 1$) from (4), (5), and (12). The shapes of the remaining five basic functions of FPs are shown in Fig. 5.

To change the shape of each component, a random shape function is chosen from the eight basic functions for each environment. In FPs, contrary to MPB, DF1, and GPB, components do not have a width parameter, and their gradients are determined based on their basic functions. In addition, the sizes and locations of hypercubes change over time, which leads to changing the size of components. Moreover, component locations inside their respective hypercubes and their height change over time using a random dynamic step. In benchmarks, such as FPs, whose basin of attractions of components is determined by dividing the landscape into hypercubes, there is no invisible component ($m = m'$). In FPs, a dynamic for changing the number of components/hypercubes is used to cover problems in which the number of components changes over time.

Several transformations are used in FPs. An important and related to single-objective DOP transformation of FPs is "setup of dependencies," which determines the variable interactions by redefining the distance in the eight basic functions. However, the study did not provide any analysis to show the variable interactions when the Euclidean distance (default distance) is used in comparison to the redefined version. According to our analysis using DG2 [60], although it is claimed that the redefinition of distance transformation can determine variable interactions and produce partially separable components, it only can generate fully separable and fully nonseparable cases. However, it should be mentioned that this can potentially change the intensity of variable interactions and thus affects the hardness of the problem for optimizers. Furthermore, the redefinition of distance transformation can only affect the complexity of exploring components inside their hypercubes and it is not capable of changing the separability of the search space. In fact, the generated landscapes by FPs are fully nonseparable when there is more than one component. Another transformation used in FPs is "setup of domino convergence" in which the contribution of each dimension will be different from other dimensions by redefining distance in the eight basic functions. Moreover, two transformations from [20] are used in FPs to add smooth irregularities (without changing modality condition) and symmetry breaking.

C. Popularity of DOP Benchmarks

In this part, we investigate the popularity of DOP benchmarks in the literature. The search has been conducted using different engines, including Scopus, Google Scholar, ScienceDirect, IEEE Xplore, ACM Digital Library, and Springer. As the searching keywords, we used different terms, such as DOPs, dynamic environments, evolutionary dynamic optimization, and uncertain environments. Then, we have selected references in which DOP algorithms/frameworks were tested on the unconstrained single-objective continuous DOP benchmarks since 1999. This process was done in August 2019, and 176 journal papers, book chapters, and conference papers have been selected for our investigation.

Fig. 6 illustrates the distribution of the DOP benchmarks utilization. In this part, we have categorized DOP benchmarks into five groups: 1) MPB and its modified versions; 2) GDBG [28], [55], [56] which contains all versions of CDBG and RDBG; 3) DF1; 4) shifting environments in which shift functions from MP [26] are utilized for moving static functions, such as Sphere, Rastrigin, Griewank, Ackley, and Rosenbrock; and 5) other DOP benchmarks. If in a reference where more than one of the above groups of benchmarks is used, we have considered them in the counter of each used group. As can be seen in Fig. 6, MPB and its modified versions are the most commonly used benchmark in continuous DOP.

D. Discussion

Table I summarizes the characteristics of problems and components that are generated by each reviewed DOP benchmark in this section alongside the proposed GMPB. As can be seen,

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on October 19,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CHARACTERISTICS OF THE REVIEWED DOP BENCHMARKS IN SECTION II AND THE PROPOSED GMPB

	Characteristic	DOP benchmark														
		AB[b]	MP	DR[c]	DSS	MPB	DF1	GPB	MVB	RDBG	CDBG[c]	IAC[c]	HDMPB	CMPB	FPs	GMPB
Landscape properties	Controllable number of components	×	×	×	✓	✓	✓	✓	✓	✓	×	×	✓	✓	✓	✓
	Modularity	×	×	×	×	×	×	×	×	×	✓	✓	✓	✓	×	✓
	Heterogeneity	×	×	×	×	×	×	×	×	×	✓	✓	×	✓	×	✓
	Imbalance	×	×	×	×	×	×	×	×	×	✓	✓	×	✓	×	✓
Component properties	Separable[a]	-	✓	-	✓	✓	✓	✓	✓	✓	-	-	✓	✓	✓	✓
	Non-separable[a]	-	×	-	×	×	×	×	×	×	-	-	×	×	✓	✓
	Ill-conditioning	-	×	-	×	×	×	×	×	×	-	-	×	×	✓	✓
	Smooth	-	✓	-	✓	✓	✓	✓	✓	✓	-	-	✓	✓	✓	✓
	Irregular	-	×	-	×	×	×	×	×	×	-	-	×	×	✓	✓
	Unimodal	-	✓	-	✓	✓	✓	✓	✓	✓	-	-	✓	✓	✓	✓
	Multimodal	-	×	-	×	×	×	×	×	×	-	-	×	×	×	✓
	Symmetric	-	✓	-	✓	✓	✓	✓	✓	✓	-	-	✓	✓	✓	✓
	Asymmetric	-	×	-	×	×	×	×	×	×	-	-	×	×	✓	✓
	Basin of attraction[d]	-	-	-	HC	MF	MF	MF	MF[e]	MF	-	-	MF	MF	HC	MF
Change	Location of components	✓	✓	×	×	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Size of components	×	×	×	✓	✓	✓	✓	✓	✓	✓	×	✓	✓	✓	✓
	Shapes of components	✓	×	×	×	×	×	×	×	×	×	✓	×	×	✓	✓
	Rotation	×	×	✓	×	×	×	×	×	×[f]	✓	✓	×	×	×	✓
	Number of visible optima	✓	×	×	×	✓	✓	✓	✓	✓	×	×	✓	✓	✓	✓
	Condition number of components	×	×	×	×	×	×	×	×	×	×	×	×	×	×	✓
	Complexity of components	×	×	×	×	×	×	×	×	×	×	✓	×	×	×	✓

[a] Determined according to empirical analysis on a dimension-wise manner.
[b] Component formula has not been provided in [25].
[c] DR, CDBG, and IAC do not use distance-based component functions to produce multiple components on the landscape.
[d] Hypercube (HC) and $\max(\cdot)$ function (MF) based basin of attractions.
[e] In MVB, a $\min(\cdot)$ function is used to determine the basin of attractions of components.
[f] The rotation is used to relocate components' centers.

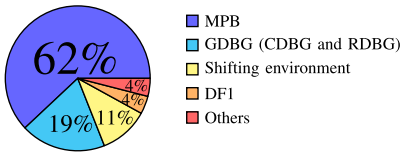


Fig. 6. Distribution of the DOP benchmarks utilization since 1999.

although all the previous DOP benchmarks can generate some problem characteristics, we cannot find a benchmark generator exhibiting different combinations of problem characteristics. One important limitation of the previous benchmarks is that each of their components consists of a single peak, which is easy to optimize because they are smooth, symmetric, and unimodal, which may not be the case of many real-world problems.

According to Table I, among all previous DOP benchmarks, the state-of-the-art FPs is capable of generating components with a variety of problem properties by using some transformations. However, it still has some considerable limitations. FPs is not easy to understand and hard to implement. In addition, some of its transformations cannot be used simultaneously (such as setup of dependencies and domino convergence). Moreover, ill-conditioning, irregularities, and symmetry breaking transformations are static. Furthermore, components generated in FPs are unimodal because the transformations do not change local modality. Finally, FPs cannot produce modular, heterogeneous, and imbalanced problems. In terms of modularity, CMPB [18] is the only benchmark capable of generating modular, heterogeneous, and imbalanced problems. In CMPB, an automated weight is assigned to each subfunction to avoid the situations in which partially separable subfunctions become dominated by fully separable dimensions in terms of contribution to overall fitness value. However, this circumstance can happen when there are several low-dimensional (such as 2-D and 3-D) subfunctions. Therefore, CMPB is not suitable to generate partially separable problem instances whose number of low-dimensional subfunctions are more than the ones with higher dimensions. In the next section, GMPB is introduced, which addresses all the above issues.

III. GENERALIZED MOVING PEAKS BENCHMARK

In this section, the details of the GMPB are given. We start with the baseline of the traditional MPB (5) and a series of modifications are proposed in order to incorporate the following features into the new design.

1) Modularity: Most DOP benchmarks whose generated landscape contains multiple components (such as MPB, FPs, and DF1 whose components are single local peaks) are non-separable in nature [18]; therefore, they are not capable of generating modular test instances. This limitation motivates the design of a new benchmark capable of producing modular test instances with heterogeneity and imbalance characteristics. In GMPB, subfunctions can vary in size, shape, change intensity, and contribution to the overall fitness value.

2) Component Complexity: Although most DOP benchmarks have a multimodal landscape, each component is unimodal, smooth, regular, separable, and symmetric, which makes them easy to optimize. GMPB is capable of generating a wide range of easy to hard to optimize components by adding the following properties to each component with varying degree of intensity:

a) Ill-conditioning: Ill-conditioning is an important property of many real-world problems [20], [24], which cannot

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on October 19,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

be found in the components generated by most DOP benchmarks due to the circular nature of the level curves of each peak. This motivates the design of a new benchmark capable of generating ill-conditioned peaks whose condition number *changes over time*.

b) Asymmetry: In almost all DOP benchmarks, components are symmetric which is undesirable for the following reasons: 1) symmetry of the peaks may be in favor of some special operators that use Gaussian distributions to generate new solutions [20], [24] and 2) they are easy to optimize on a dimension-wise basis due to their symmetric nature. The GMPB is capable of generating symmetric to highly asymmetric components.

c) Component separability: In addition, as shown analytically [18], an n-dimensional peak generated by MPB and DF1 by (5) is separable which makes it easier to optimize. As demonstrated in Table I, most other DOP benchmarks generate components (peaks) that can be optimized in a dimension-wise manner. These empirical observations suggest that they are separable. Rotating a component is a way of changing its variable interaction and making it *fully nonseparable*, which is more difficult to optimize [20], [24], [61]. In GMPB, components can be rotated using rotation matrices to have different degrees of variable interactions. Using rotation matrices to transform problem spaces has been used in many benchmarks for static [20], [22], [24], [62] and dynamic [4], [27], [28], [56], [63] optimization problems. Moreover, by rotating a component over time, the degree of variable interaction changes dynamically. In the remainder of this section, the procedure of transforming MPB into GMPB is described.

d) Component modality: As mentioned in Table I, generated components of all existing benchmarks are unimodal which makes them easy to optimize. Components in GMPB can be unimodal to highly multimodal.

A. Components With Varying Condition Number

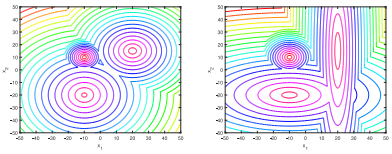
According to (5), the width of each component (peak) is the same in all dimensions, which makes the shape of components cone like with circular contour lines [see Fig. 7(a)]. To alleviate this, the width of a component is changed from a scalar variable to a d-dimensional vector. We know that the Euclidean distance can be shown as the square root of a dot product, that is, $\|\mathbf{a}\| = \sqrt{\mathbf{a}^T\mathbf{a}}$. Therefore, MPB's baseline function in (5) can be rewritten as follows:

$$f^{(t)}(\mathbf{x}) = \max_{i\in\{1,\ldots,m\}} \{h_i^{(t)} - \sqrt{\hat{w}_i^{(t)}(\mathbf{x}-\mathbf{c}_i^{(t)})^\top(\mathbf{x}-\mathbf{c}_i^{(t)})}\} \quad (20)$$

where $\sqrt{\hat{w}_i^{(t)}} = w_i^{(t)}$. Instead of factoring the width, we can represent it as a diagonal matrix $\mathbf{W}_i^{(t)} = \hat{w}_i^{(t)}\mathbf{I}$

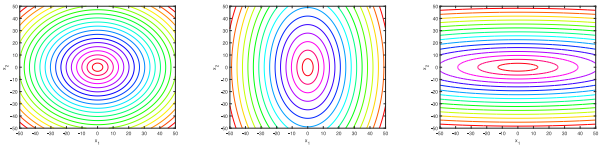
$$f^{(t)}(\mathbf{x}) = \max_{i\in\{1,\ldots,m\}} \{h_i^{(t)} - \sqrt{(\mathbf{x}-\mathbf{c}_i^{(t)})^\top\mathbf{W}_i^{(t)}(\mathbf{x}-\mathbf{c}_i^{(t)})}\}. \quad (21)$$

By allowing the main diagonal elements of $\mathbf{W}$ to be set arbitrarily, the contour lines of the function become ellipses which are aligned with the coordinate axes. Fig. 7(b) shows an example of the MPB made using (21) with three components. One of the components (the one with circular contour) has the same width in each direction which makes it similar to the components made by (5). The other two components have different width values for each dimension resulting in elliptical contour lines.



(a) (b)

Fig. 7. Comparing two landscapes with and without ill-conditioning in components. (a) Standard MPB generated by (5). (b) Ill-conditioned MPB generated by (21) .



(a) (b) (c)

Fig. 8. Three different 2-D components with different ill-conditioning degrees. The values of widths (w) of each peak and their corresponding condition number (c) are shown. (a) $w = [1, 1]$ and $c = 1$. (b) $w = [2, 4]$ and $c = 2.2$ (c) $w = [8, 2]$ and $c = 4$.

By calculating the condition number of the diagonal matrix $\mathbf{W}$, the ill-conditioning degree of each component can be determined. The condition number of a matrix is generally calculated through singular value decomposition of the matrix and finding the ratio between the largest and smallest diagonal elements of its singular value matrix. Since $\mathbf{W}$ is a diagonal matrix, its singular value matrix has the same diagonal values but in a sorted order; therefore, the condition number of $\mathbf{W}$ can be calculated without any singular decomposition. Since the diagonal values of $\mathbf{W}$ are squared values of the width vector, the condition number is equal to the squared condition number of the component. Consequently, the condition number of the generated hyperellipsoid by (21) is the square root of the condition value of $\mathbf{W}$. If all the diagonal values of $\mathbf{W}$ are equal, then the problem is the same as the original MPB by (5) and the condition number of $\mathbf{W}$ will be 1. This condition number shows that the component does not have ill-conditioning. However, when the width values are different in different dimensions, the condition number of $\mathbf{W}$ grows. From a geometric perspective, the condition number of a hyperellipsoid is the ratio of its largest diameter to its smallest one. Fig. 8 illustrates three components with different ill-conditioning degrees from the geometric point of view. As demonstrated in Fig. 8, by increasing the difference between width values, the ratio between the largest and smallest diameters of the ellipsoid component grows. Consequently, the level of ill-conditioning is increased.

B. Components With Varying Variable Interaction Degree

As mentioned before, components of the MPB are circular peaks, separable, and easy to optimize. However, even after changing the MPB baseline to (21), peaks remain separable

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on October 19,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

due to their symmetric nature. To change the principal axes of the elliptic contour lines, an orthonormal matrix $\mathbf{R}$ can be included as follows:

$$f^{(t)}(\mathbf{x}) = \max_{i\in\{1,\ldots,m\}}\left\{h_i^{(t)} - \sqrt{\left(\mathbf{x}-\mathbf{c}_i^{(t)}\right)^{\top}{\mathbf{R}_i^{(t)}}^{\top}\mathbf{W}_i^{(t)}\mathbf{R}_i^{(t)}\left(\mathbf{x}-\mathbf{c}_i^{(t)}\right)}\right\} \tag{22}$$

where $\mathbf{R}_i^{(t)}$ is the rotation matrix of component i in the t-th environment. Equation (22) is capable of generating components with elliptic contour lines whose principal axes are defined by the rotation matrix $\mathbf{R}$. In (22), the ill-conditioning intensity of a component can be obtained by calculating the condition number of $\mathbf{R}^{\top}\mathbf{W}\mathbf{R}$ which is equal to the condition number of $\mathbf{W}$ because $\mathbf{R}$ is an orthonormal matrix and does not change the condition number of $\mathbf{W}$ after multiplication. Contrary to the components generated by (5) and (21), each generated component i by (22) is fully nonseparable whose degree of variable interactions depends on $\mathbf{R}_i$. Fig. 9(a) shows the landscape generated by (22) with three components [same component centroids as in Fig. 7(b)].

The rotation matrix $\mathbf{R}_i$ is obtained by rotating the projection of $\mathbf{x}$ onto all x_p–x_q planes by a given angle θ. The total number of unique planes which will be rotated is $\binom{d}{2} = ([d(d-1)]/2)$. For rotating each x_p–x_q plane by a certain angle (θ), a *Givens rotation matrix* $\mathbf{G}_{(p,q)}$ must be constructed. To this end, first, $\mathbf{G}_{(p,q)}$ is initialized to an identity matrix $\mathbf{I}_{d\times d}$; then, four elements of $\mathbf{G}_{(p,q)}$ are altered as

$$\mathbf{G}_{(p,q)}(p,p) = \cos\left(\theta^{(t)}\right) \tag{23}$$

$$\mathbf{G}_{(p,q)}(q,q) = \cos\left(\theta^{(t)}\right) \tag{24}$$

$$\mathbf{G}_{(p,q)}(p,q) = -\sin\left(\theta^{(t)}\right) \tag{25}$$

$$\mathbf{G}_{(p,q)}(q,p) = \sin\left(\theta^{(t)}\right) \tag{26}$$

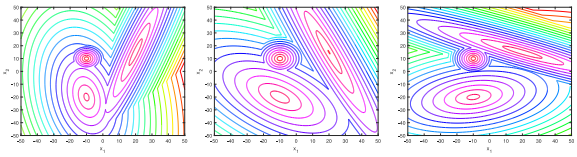
where $\mathbf{G}_{(p,q)}(i,j)$ is the element at the ith row and the jth column of $\mathbf{G}_{(p,q)}$. $\mathbf{R}_i$ in the t-th environment is calculated by

$$\mathbf{R}_i^{(t)} = \prod_{(p,q)\in\mathcal{P}} \mathbf{G}_{(p,q)} \times \mathbf{R}_i^{(t-1)} \tag{27}$$

where $\mathcal{P}$ contains all unique pairs of dimensions defining all possible planes in a d-dimensional space. The order of the multiplications of the *Givens rotation matrices* is random. The reason behind using (27) for calculating $\mathbf{R}$ is that we aim to have control on the rotation matrix based on an angle severity $\tilde{\theta}$. Note that the initial $\mathbf{R}_i^{(0)}$ for problem instances with rotation property is obtained by using the Gram–Schmidt orthogonalization method on a matrix with normally distributed entries. Fig. 9 shows an example of component rotation in the proposed benchmark in which by changing the angle of rotation, the degree of variable interaction changes.

C. Asymmetric Components With Irregularities

Although optimizing a rotated component is harder for many algorithms due to its nonseparable nature [61], each component can still be considered *easy to optimize* since the components generated by (22) are smooth, regular, and unimodal. To add irregularity and local optima to the components, a transformation function is added to (22)



(a) (b) (c)

Fig. 9. Clockwise rotating a landscape generated by (27) in 2-D space. (a) Landscape generated by (22). (b) After rotating 45°. (c) After rotating 90°.

$$f^{(t)}(\mathbf{x}) = \max_{i\in\{1,\ldots,m\}}\left\{h_i^{(t)} - \sqrt{\mathbb{T}\left(\left(\mathbf{x}-\mathbf{c}_i^{(t)}\right)^{\top}\mathbf{R}^{(t)\top}, i\right)\mathbf{W}^{(t)}\mathbb{T}\left(\mathbf{R}^{(t)}\left(\mathbf{x}-\mathbf{c}_i^{(t)}\right), i\right)}\right\} \tag{28}$$

where $\mathbb{T}(\mathbf{y}, i) : \mathbb{R}^d \mapsto \mathbb{R}^d$ is calculated as

$$\mathbb{T}(y_j, i) = \begin{cases} \exp\left(\log(y_j) + \tau_i^{(t)}\left(\sin\left(\eta_{i,1}^{(t)}\log(y_j)\right)\right.\right. & \text{if } y_j > 0 \\ \left.\left. + \sin\left(\eta_{i,2}^{(t)}\log(y_j)\right)\right)\right) & \\ 0 & \text{if } y_j = 0 \\ -\exp\left(\log(|y_j|) + \tau_i^{(t)}\left(\sin\left(\eta_{i,3}^{(t)}\log(|y_j|)\right)\right.\right. & \\ \left.\left. + \sin\left(\eta_{i,4}^{(t)}\log(|y_j|)\right)\right)\right) & \text{if } y_j < 0 \end{cases} \tag{29}$$

where y_j is the jth dimension of $\mathbf{y}$, and $\tau_i^{(t)}$ and $\eta_{i,k}^{(t)}$ for $k = 1,\ldots,4$ are five dynamic parameters that determine the intensity of irregularities and modality of the ith component. Equation (28) is designed in a way that the $\mathbf{c}$ positions have the best fitness values among all optima in their component and the global optimum is still $\mathbf{c}$ of the component with the largest height. In GMPB, the term ith *component* means the area containing all the points whose Euclidean distances to the $\mathbf{c}_i$ is less than those to all $\mathbf{c}_j$ where $i \neq j$.

Fig. 10 shows the effect of the different values of τ and $\eta_{1,2,3,4}$ on the shape of a component from Fig. 10(a). As can be seen, higher values of τ and $\eta_{1,2,3,4}$ increase the irregularities and number of local optima in the component. In addition, according to 10(b) and (c), when all four values of $\eta_{1,2,3,4}$ are equal, the component shape is symmetric. On the other hand, according to Fig. 10(d)–(f), when $\eta_{1,2,3,4}$ are unequal, the component shape is asymmetric. By changing τ and $\eta_{1,2,3,4}$ over time for each component, their intensity of irregularities, number of local optima, and asymmetry degree change over time. Fig. 11 shows an example for how the shape of a component changes as the values of τ and $\eta_{1,2,3,4}$ change over time. Furthermore, Fig. 12 shows examples of landscapes with the components from Figs. 7(a) and 9(c) that are made irregular by different values of τ and $\eta_{1,2,3,4}$ for each peak. Equation (28) is the baseline of GMPB.

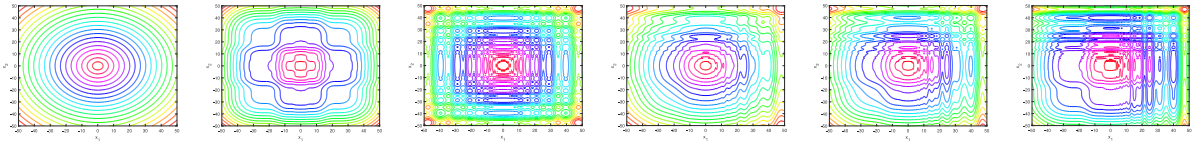
D. Modularity, Heterogeneity, and Imbalance

The modularity can be obtained by composing several GMPB

$$F^{(t)}(\mathbf{x}) = \sum_{i=1}^{k} f_i^{(t)}(\mathbf{x}) \tag{30}$$

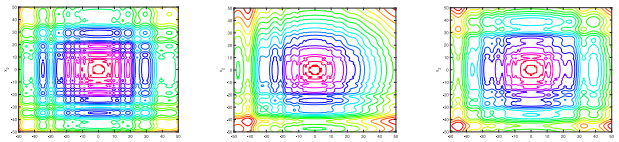
where $f_i^{(t)}$ is the ith baseline, and k is the number of subfunctions in the composition function. Since the baseline of GMPB

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on October 19,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.



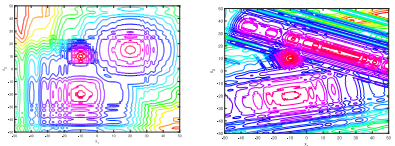
(a) (b) (c) (d) (e) (f)

Fig. 10. Illustrating the effect of different values of τ and $\eta_{1,2,3,4}$ in (29) on a component. (a) $\tau = 0$, $\eta_{1,2,3,4} = 0$. (b) $\tau = 0.1$, $\eta_{1,2,3,4} = 5$. (c) $\tau = 0.1$, $\eta_{1,2,3,4} = 20$. (d) $\tau = 0.05$, $\eta_1 = 0$, $\eta_2 = 5$, $\eta_3 = 15$, and $\eta_4 = 25$. (e) $\tau = 0.1$, $\eta_1 = 0$, $\eta_2 = 5$, $\eta_3 = 15$, and $\eta_4 = 25$. (f) $\tau = 0.2$, $\eta_1 = 0$, $\eta_2 = 5$, $\eta_3 = 15$, and $\eta_4 = 25$.



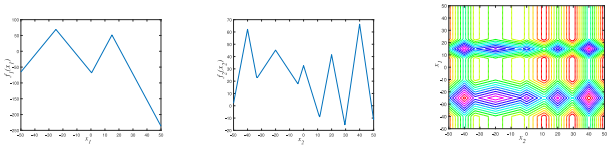
(a) (b) (c)

Fig. 11. Illustrating the effect of dynamic τ and $\eta_{1,2,3,4}$ on shape of a component. (a) First environment. (b) Second environment. (c) Third environment.



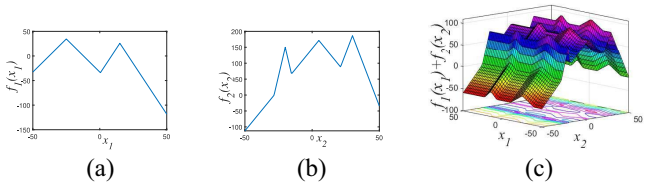
(a) (b)

Fig. 12. Transforming a landscape with three components using (29) with $\tau \in \mathcal{U}[0, 0.2]$ and $\eta_{1,2,3,4} \in \mathcal{U}[0, 25]$ for each main peak. (a) Transformed version of the landscape from Fig. 7(a). (b) Transformed version of the landscape from Fig. 9(c).



(a) (b) (c)

Fig. 13. Exponentially growing number of optima by composing subfunctions in (30). (a) 1-D subfunction with two optima. (b) 1-D subfunction with five optima. (c) 2-D modular landscape with ten optima by composing (a) and (b).



(a) (b) (c)

Fig. 14. Imbalanced landscape generated by composing two subfunctions with different weights ω in (32). (a) 1-D subfunction with $\omega = 0.5$. (b) 1-D subfunction with $\omega = 3$. (c) 2-D imbalanced landscape by composing (a) and (b).

is nonseparable, 1-D subfunctions will be used for generating fully separable dimensions. Each subfunction in (30) can have a different number of components, dimensions, and/or change intensities. Consequently, since each subfunction can have different landscapes with different features and challenges, the generated problems by GMPB can possess the heterogeneity characteristic. A natural result of composing GMPB baselines in (30) is an exponential growth in the total number of optima in the landscape that can turn to the global optimum after environmental changes. To demonstrate this challenging characteristic, we provide an illustrative example in Fig 13. In Fig. 13(a) and (b), two 1-D landscapes with 2 and 5 optima, respectively, are shown (for simplicity, τ and $\eta_{1,2,3,4}$ are set to zero). The 2-D modular landscape constructed by (30) results in a total of ten optima as shown in Fig. 13(c). To be specific, the number of optima in the modular landscapes constructed by (30) is equal to a product of the numbers of optima in the subfunctions.

According to the nature of baseline in (28), the fitness value is independent of the dimension number, and it is dependent on the highest h. Therefore, there is a balance between the contribution of subfunctions f in (30), which is not common in many real-world problems [3], [24], [64]. To address this issue, the contribution of each MPB is normalized according to its number of dimensions relative to the dimension of the problem

$$F^{(t)}(\mathbf{x}) = d^{-1} \sum_{i=1}^{k} d_i f_i^{(t)}(\mathbf{x}) \tag{31}$$

where d is the dimension of the problem, and d_i is the dimension of the subfunction f_i. By using (31), the shortcoming of CMPB [18] regarding dominating contribution of higher dimensional subfunctions by low-dimensional ones is addressed. After normalizing the contribution of each subfunction, an imbalance coefficient ω will be added to the composition, which can be utilized to generate different imbalance patterns

$$F^{(t)}(\mathbf{x}) = d^{-1} \sum_{i=1}^{k} \omega_i d_i f_i^{(t)}(\mathbf{x}) \tag{32}$$

where ω_i controls the contribution of subfunction f_i for generating imbalance. In an imbalanced problem, the contributions of subfunctions on the overall fitness value are different. Consequently, some subfunctions become more significant for optimizers. Fig. 14(c) illustrates a 2-D modular landscape constructed by composing two 1-D subfunctions shown in Fig. 14(a) and (b). The range of fitness values in the subfunction from Fig. 14(b) is considerably larger than that of the subfunction from Fig. 14(a). Consequently, the composed problem shown in Fig. 14(c) is an imbalanced problem.

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on October 19,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

According to the illustrated landscape in Fig. 14(c), the progress of optimization by moving toward the direction of x_2 axis is more important than moving in x_1.

E. Dynamics

For each subfunction f_i in (32), the height, width vector, angle, irregularity parameters in (29), and the center of the jth component change from one environment to the next according to the following update rules:

$$\mathbf{c}_{i,j}^{(t+1)} = \mathbf{c}_{i,j}^{(t)} + \tilde{s}_i \frac{\mathbf{r}}{\|\mathbf{r}\|} \tag{33}$$

$$h_{i,j}^{(t+1)} = h_{i,j}^{(t)} + \tilde{h}_i \mathcal{N}(0,1) \tag{34}$$

$$w_{i,j,k}^{(t+1)} = w_{i,j,k}^{(t)} + \tilde{w}_i \mathcal{N}(0,1) \tag{35}$$

$$\theta_{i,j}^{(t+1)} = \theta_{i,j}^{(t)} + \tilde{\theta}_i \mathcal{N}(0,1) \tag{36}$$

$$\eta_{i,j,l}^{(t+1)} = \eta_{i,j,l}^{(t)} + \tilde{\eta}_i \mathcal{N}(0,1), l \in \{1,2,3,4\} \tag{37}$$

$$\tau_{i,j}^{(t+1)} = \tau_{i,j}^{(t)} + \tilde{\tau}_i \mathcal{N}(0,1) \tag{38}$$

where $\mathcal{N}(0,1)$ is a random number drawn from a Gaussian distribution with mean 0 and variance 1, $\mathbf{r}$ is a vector of randomly generated numbers by $\mathcal{N}(0,1)$, $\tilde{h}_i$, $\tilde{w}_i$, $\tilde{s}_i$, $\tilde{\theta}_i$, $\tilde{\eta}_i$, and $\tilde{\tau}_i$ are height, width, shift, angle, and irregularity parameters' change severities of components in the ith subfunction, respectively. $w_{i,j,k}$ shows the width of the jth component in the kth dimension of the ith subfunction. In addition, $h_{i,j}$, $\theta_{i,j}$, and $\mathbf{c}_{i,j}$ show the height, angle, and position of the jth component in the ith subfunction f, respectively.

Outputs of (33)–(38) are bounded as follows: $h_{i,j} \in [h_{\min}, h_{\max}]$, $w_{i,j,k} \in [w_{\min}, w_{\max}]$, $\mathbf{c}_{i,j} \in [Lb, Ub]^d$, $\tau \in [\tau_{\min}, \tau_{\max}]$ and $\eta_{1,2,3,4} \in [\eta_{\min}, \eta_{\max}]$, and $\theta_{i,j} \in [\theta_{\min}, \theta_{\max}]$, where Lb and Ub are maximum and minimum problem space bounds. For keeping the above-mentioned values in their bounds, a *reflect* method is utilized. Assume $a^{(t+1)} = a^{(t)} + b$ represents one of the equations (34)–(38). Then, the output based on the reflect method is

$$a^{(t+1)} = \begin{cases} a^{(t)} + b, & \text{if } a^{(t)} + b \in [a_{\min}, a_{\max}] \\ 2 \times a_{\min} - a^{(t)} - b, & \text{if } a^{(t)} + b < a_{\min} \\ 2 \times a_{\max} - a^{(t)} - b, & \text{if } a^{(t)} + b > a_{\max}. \end{cases} \tag{39}$$

The dynamics which are used for GMPB in (34)–(38) are based on the *random step*, which is also used in MPB. Any dynamics in (14)–(19) can also be used in GMPB. For example, researchers who are interested in periodical environmental changes can use the pendulum-based environmental changes similar to (11), or use the cyclic center relocations from [57]. To have predictable component relocations, (33) can be replaced with (8) and (9) (set $\lambda = 1$). To have undetectable environmental changes, a simple noise generator can be added to the fitness function. In addition, inspired by (10), the β threshold can be added to the benchmark to create vast flat regions whose fitness remains unchanged in environmental changes. By applying the aforementioned modifications, most change detection methods that work by re-evaluating solutions [2] cannot work properly. However, it should be mentioned that in most real-world DOPs, the algorithm will be informed about the environmental changes, and it is not necessary to use a change detection mechanism [1], [16]. Moreover, having partial environmental changes can be obtained by assigning different change frequencies to different subfunctions or different components. This can be done using a simple probability-based method in which when an environmental change happens, components or subfunctions with a predefined probability can be involved in change. Finally, dynamic constraints can be easily added to GMPB. In Section S-III of the supplementary document, we have provide a discussion on the existing dynamic-constrained benchmarks that can be incorporated to the GMPB.

TABLE II
SUMMARY OF SCENARIOS' CHARACTERISTICS, INCLUDING SEPARABILITY [FULLY NONSEPARABLE (N) AND PARTIALLY SEPARABLE (P)], COMPONENT MODALITY [MULTIMODAL (M) AND UNIMODAL (U)] AND ILL-CONDITIONING PLUS ROTATION

Feature	Scenario							
	f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8
Separability	N	N	N	N	P	P	P	P
Component modality	U	U	M	M	U	U	M	M
Ill-conditioning and rotation	×	✓	×	✓	×	✓	×	✓

F. Proposed Scenarios

We propose eight problem instances exhibiting different combinations of problem characteristics using the GMPB benchmark generator. To design these scenarios, the characteristics of components and problems are categorized into three groups: separability, local-modality, and ill-conditioning/rotation. The first group is defined according to the modularity of the problems based on (32). Two groups of problems are designed according to separability, including fully nonseparable and partially separable ones. For component modality, problems are categorized based on the value assigned to τ and $\eta_{1,2,3,4}$. If the values are set to zero, components will be smooth, regular, and unimodal. Otherwise, the problem contains components with different shapes and various features. Finally, if the rotation matrices are set to the identity matrix, angle severity is set to zero, and the width vector has the same value on all dimensions (with the same random number for all dimensions at environmental changes), components will have circular contour and become fully separable. Else, components will be ill-conditioned with dynamic condition number and variable interactions. The properties of the eight scenarios and the categories they belong to are summarized in Table II. The parameter settings of the problem instances and component characteristics are shown in Table III. To investigate the efficiency of DOP algorithms on each scenario with different levels of difficulty, two different parameter settings are considered for change frequency (ϑ), shift severity ($\tilde{s}$), and the number of components (m), which are shown in Table IV. The first being the default settings while the second group generates more challenging scenarios.

G. Discussion

To study the effect of different component characteristics, several well-known optimizers are selected and tested

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on October 19,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

TABLE III
PARAMETER SETTINGS (PART 1) OF SCENARIOS f_1–f_8 FROM TABLE II

Parameter	Symbol	Scenario							
		f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8
Number of subfunctions	k	1				6			
Dimension	d	10							
Number of fully separable dimensions	d'	0				2			
Number of Nonseparable subfunctions	d''	$\{10\}$				$\{4, 2, 2\}$			
Angle severities	$\tilde{\theta}_i$	0	$\pi/9$	0	$\pi/9$	0	$\mathcal{U}[\pi/12, \pi/6]$	0	$\mathcal{U}[\pi/12, \pi/6]$
Height severities	$\tilde{h}_i$	7				$\mathcal{U}[5, 9]$			
Width severities	$\tilde{w}_i$	1				$\mathcal{U}[0.5, 1.5]$			
Irregularity parameter τ severities	$\tilde{\tau}_i$	0		0.05		0		$\mathcal{U}[0.025, 0.075]$	
Irregularity parameter η severities	$\tilde{\eta}_i$	0		2		0		$\mathcal{U}[1, 3]$	
Weight of subfunction i	ω_i	1				$\mathcal{U}[0.5, 3]$			
Search range	$[Lb, Ub]^d$	$[-50, 50]^d$							
Height range	$[h_{\min}, h_{\max}]$	$[30, 70]$							
Width range	$[w_{\min}, w_{\max}]^d$	$[1, 12]^d$							
Angle range	$[\theta_{\min}, \theta_{\max}]$	-	$[-\pi, \pi]$	-	$[-\pi, \pi]$	-	$[-\pi, \pi]$	-	$[-\pi, \pi]$
Irregularity parameter τ range	$[\tau_{\min}, \tau_{\max}]$	-	-	$[0, 0.4]$	$[0, 0.4]$	-	-	$[0, 0.4]$	$[0, 0.4]$
Irregularity parameter η range	$[\eta_{\min}, \eta_{\max}]$	-	-	$[10, 25]$	$[10, 25]$	-	-	$[10, 25]$	$[10, 25]$
Number of Environments	T	100							

TABLE IV
TWO GROUPS OF PARAMETER SETTINGS (PART 2) OF SCENARIOS f_1–f_8

Parameter	Symbol	Default setting		Challenging setting	
		f_{1-4}	f_{5-8}	f_{1-4}	f_{5-8}
Shift severities	$\tilde{s}_i$	2	$\mathcal{U}[1, 3]$	4	$\mathcal{U}[3, 5]$
Numbers of components in each subfunction i	m_i	10	$\mathcal{U}[5, 15]$	25	$\mathcal{U}[15, 35]$
Change frequency	ϑ	5000	5000	2500	2500

on the generated components by GMPB. The results and the correspondence analysis can be found in Sections S-II-C1 and S-II-C2 of the supplementary document. To investigate the performance of existing DOP algorithms on the proposed GMPB scenarios, a set of 11 different DOP algorithms is chosen (see Section S-II-B of the supplementary document). The experimental results on the eight GMPB scenarios can be found in Sections S-II-D and S-II-E of the supplementary document. The experimental results clearly indicate the poor efficiencies of the existing algorithms in handling the challenges posed by GMPB.

IV. CONCLUSION

In this article, we have presented a comprehensive review of continuous single-objective unconstrained DOP benchmarks. The critical review of the existing benchmark suites showed that the landscape of most well-known DOP benchmarks is made up of one or more smooth, separable, symmetric, unimodal components with neutral condition numbers, which are easy to optimize and not representative of many real-world problems. We also have observed that there is no DOP benchmark in the literature capable of generating problems with different combinations of characteristics. To address the above-mentioned shortcomings, we have proposed a new GMPB. GMPB is capable of generating problems and components with varieties of properties. GMPB is configurable and can generate problems ranging from simple unimodal, smooth, regular, separable, symmetric, balanced, and homogeneous to highly multimodal, imbalanced, heterogeneous, and partially separable problems with ill-conditioned, highly asymmetric, irregular, and nonseparable components. Moreover, all the aforementioned features are dynamic and change over time in GMPB. Several well-known optimizers and DOP algorithms have been used to investigate the performance of the existing algorithms on the generated problem instances by GMPB. The experimental results have indicated the poor efficiencies of the existing algorithms in handling the challenges posed by GMPB. In the future, some other important characteristics can also be considered to be added to GMPB, such as dynamic constraints or multiple conflicting objectives, which are important classes of DOPs [10]–[12]. In addition, due to the importance of the discrete and combinatorial DOPs, investing existing problems and benchmark generators in this domain is an important topic for future work [6], [7].

REFERENCES

[1] T. T. Nguyen, "Continuous dynamic optimisation using evolutionary algorithms," Ph.D. dissertation, Elect. Electron. Eng., Univ. Birmingham, Birmingham, U.K., 2011.

[2] T. T. Nguyen, S. Yang, and J. Branke, "Evolutionary dynamic optimization: A survey of the state of the art," *Swarm Evol. Comput.*, vol. 6, pp. 1–24, Oct. 2012.

[3] D. Yazdani, "Particle swarm optimization for dynamically changing environments with particular focus on scalability and switching cost," Ph.D. dissertation, Dept. Comput. Sci., Liverpool John Moores Univ., Liverpool, U.K., 2018.

[4] C. Li, T. T. Nguyen, S. Zeng, M. Yang, and M. Wu, "An open framework for constructing continuous optimization problems," *IEEE Trans. Cybern.*, vol. 49, no. 6, pp. 2316–2330, Jun. 2019.

[5] C. Cruz, J. R. González, and D. A. Pelta, "Optimization in dynamic environments: A survey on problems, methods and measures," *Soft Comput.*, vol. 15, no. 7, pp. 1427–1448, 2011.

[6] S. Yang, Y. Jiang, and T. T. Nguyen, "Metaheuristics for dynamic combinatorial optimization problems," *IMA J. Manag. Math.*, vol. 24, no. 4, pp. 451–480, 2013.

[7] M. Mavrovouniotis, F. M. Muller, and S. Yang, "Ant colony optimization with local search for dynamic traveling salesman problems," *IEEE Trans. Cybern.*, vol. 47, no. 7, pp. 1743–1756, Jun. 2017.

[8] J. Branke, "Memory enhanced evolutionary algorithms for changing optimization problems," in *Proc. IEEE Congr. Evol. Comput.*, 1999, pp. 1875–1882.

[9] S. B. Gee, K. C. Tan, and H. A. Abbass, "A benchmark test suite for dynamic evolutionary multiobjective optimization," *IEEE Trans. Cybern.*, vol. 47, no. 2, pp. 461–472, Feb. 2017.

[10] S. Jiang, M. Kaiser, S. Yang, S. Kollias, and N. Krasnogor, "A scalable test suite for continuous dynamic multiobjective optimization," *IEEE Trans. Cybern.*, vol. 50, no. 6, pp. 2814–2826, Jun. 2020.

[11] S. Jiang and S. Yang, "Evolutionary dynamic multiobjective optimization: Benchmarks and algorithm comparisons," *IEEE Trans. Cybern.*, vol. 47, no. 1, pp. 198–211, Jan. 2017.

[12] T. T. Nguyen and X. Yao, "Continuous dynamic constrained optimization—The challenges," *IEEE Trans. Evol. Comput.*, vol. 16, no. 6, pp. 769–786, Dec. 2012.

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on October 19,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

[13] C. Bu, W. Luo, and L. Yue, "Continuous dynamic constrained optimization with ensemble of locating and tracking feasible regions strategies," *IEEE Trans. Evol. Comput.*, vol. 21, no. 1, pp. 14–33, Feb. 2017.

[14] Y. Wang, J. Yu, S. Yang, S. Jiang, and S. Zhao, "Evolutionary dynamic constrained optimization: Test suite construction and algorithm comparisons," *Swarm Evol. Comput.*, vol. 50, Nov. 2019, Art. no. 100559.

[15] X. Yu, Y. Jin, K. Tang, and X. Yao, "Robust optimization over time—A new perspective on dynamic optimization problems," in *Proc. IEEE Congr. Evol. Comput. (CEC)*, 2010, pp. 1–6.

[16] D. Yazdani, T. T. Nguyen, and J. Branke, "Robust optimization over time by learning problem space characteristics," *IEEE Trans. Evol. Comput.*, vol. 13, no. 1, pp. 143–155, Feb. 2019.

[17] D. Yazdani, T. T. Nguyen, J. Branke, and J. Wang, "A multi-objective time-linkage approach for dynamic optimization problems with previous-solution displacement restriction," in *Proc. Eur. Conf. Appl. Evol. Comput.*, vol. 10784, 2018, pp. 864–878.

[18] D. Yazdani, M. N. Omidvar, J. Branke, T. T. Nguyen, and X. Yao, "Scaling up dynamic optimization problems: A divide-and-conquer approach," *IEEE Trans. Evol. Comput.*, vol. 24, no. 1, pp. 1–15, Feb. 2020.

[19] W. Luo, B. Yang, C. Bu, and X. Lin, "A hybrid particle swarm optimization for high-dimensional dynamic optimization," in *Simulated Evolution and Learning* (LNCS 10593), Y. Shi *et al.*, Eds. Cham, Switzerland: Springer, 2017, pp. 981–993.

[20] N. Hansen, S. Finck, R. Ros, and A. Auger, "Real-parameter black-box optimization benchmarking 2009: Noiseless functions definitions," INRIA, New Delhi, India, Rep. RR-6829, 2010.

[21] P. N. Suganthan *et al.*, "Problem definitions and evaluation criteria for the CEC 2005 special session on real-parameter optimization," Dept. Comput. Sci., Nanyang Technol. Univ., Singapore, Rep. 2005005, 2005.

[22] J. J. Liang, P. N. Suganthan, and K. Deb, "Novel composition test functions for numerical global optimization," in *Proc. Swarm Intell. Symp.*, 2005, pp. 68–75.

[23] K. Tang, X. Li, P. N. Suganthan, Z. Yang, and T. Weise, "Benchmark functions for the CEC'2010 special session and competition on large-scale global optimization," Nat. Inspired Comput. Appl. Lab., Hefei, China, Rep., 2009.

[24] M. N. Omidvar, X. Li, and K. Tang, "Designing benchmark problems for large-scale continuous optimization," *Inf. Sci.*, vol. 316, pp. 419–436, Sep. 2015.

[25] H. G. Cobb and J. J. Grefenstette, "Genetic algorithms for tracking changing environments," in *Proc. 5th Int. Conf. Genet. Algorithms*, 1993, pp. 523–530.

[26] P. Angeline, "Tracking extrema in dynamic environments," in *Evolutionary Programming VI* (LNCS 1213), P. Angeline, R. Reynolds, J. McDonnell, and R. Eberhart, Eds. Heidelberg, Germany: Springer, 1997, pp. 335–345.

[27] K. Weicker and N. Weicker, "Dynamic rotation and partial visibility," in *Proc. Congr. Evol. Comput.*, 2000, pp. 1125–1131.

[28] C. Li and S. Yang, "A generalized approach to construct benchmark problems for dynamic optimization," in *Simulated Evolution and Learning* (LNCS 5361). Heidelberg, Germany: Springer, 2013, pp. 391–400.

[29] R. Tinos and S. Yang, "A framework for inducing artificial changes in optimization problems," *Inf. Sci.*, vol. 485, pp. 486–504, Jun. 2019.

[30] J. Branke and H. Schmeck, "Designing evolutionary algorithms for dynamic optimization problems," in *Advances in Evolutionary Computing* (Natural Computing Series), A. Ghosh and S. Tsutsui, Eds. Heidelberg, Germany: Springer, 2003, pp. 239–262.

[31] R. W. Morrison and K. A. D. Jong, "A test problem generator for non-stationary environments," in *Proc. Congr. Evol. Comput. (CEC)*, vol. 3, 1999, pp. 2047–2053.

[32] R. W. Morrison, *Designing Evolutionary Algorithms for Dynamic Environments* (Natural Computing Series). Heidelberg, Germany: Springer, 2004.

[33] T. Blackwell and J. Branke, "Multiswarms, exclusion, and anti-convergence in dynamic environments," *IEEE Trans. Evol. Comput.*, vol. 10, no. 4, pp. 459–472, Aug. 2006.

[34] R. Mendes and A. S. Mohais, "DynDE: A differential evolution for dynamic optimization problems," in *Proc. Congr. Evol. Comput.*, vol. 3, 2005, pp. 2808–2815.

[35] M. C. du Plessis and A. P. Engelbrecht, "Using competitive population evaluation in a differential evolution algorithm for dynamic environments," *Eur. J. Oper. Res.*, vol. 218, no. 1, pp. 7–20, 2012.

[36] C. Li, T. T. Nguyen, M. Yang, M. Mavrovouniotis, and S. Yang, "An adaptive multi-population framework for locating and tracking multiple optima," *IEEE Trans. Evol. Comput.*, vol. 20, no. 5, pp. 590–605, Aug. 2016.

[37] D. Yazdani, B. Nasiri, A. Sepas-Moghaddam, M. R. Meybodi, and M. Akbarzadeh-Totonchi, "mNAFSA: A novel approach for optimization in dynamic environments with global changes," *Swarm Evol. Comput.*, vol. 18, pp. 38–53, Oct. 2014.

[38] D. Yazdani, B. Nasiri, A. Sepas-Moghaddam, and M. R. Meybodi, "A novel multi-swarm algorithm for optimization in dynamic environments based on particle swarm optimization," *Appl. Soft Comput.*, vol. 13, no. 4, pp. 2144–2158, 2013.

[39] S. Yang and C. Li, "A clustering particle swarm optimizer for locating and tracking multiple optima in dynamic environments," *IEEE Trans. Evol. Comput.*, vol. 14, no. 6, pp. 959–974, Dec. 2010.

[40] C. Li and S. Yang, "A general framework of multipopulation methods with clustering in undetectable dynamic environments," *IEEE Trans. Evol. Comput.*, vol. 16, no. 4, pp. 556–577, Aug. 2012.

[41] D. Yazdani, A. Sepas-Moghaddam, A. Dehban, and N. Horta, "A novel approach for optimization in dynamic environments based on modified artificial fish swarm algorithm," *Int. J. Comput. Intell. Appl.*, vol. 15, no. 2, 10–34, 2016.

[42] H. Richter, "Detecting change in dynamic fitness landscapes," in *Proc. Congr. Evol. Comput.*, 2009, pp. 1613–1620.

[43] D. Yazdani, T. T. Nguyen, J. Branke, and J. Wang, "A new multi-swarm particle swarm optimization for robust optimization over time," in *Applications of Evolutionary Computation* (LNCS 10200), G. Squillero and K. Sim, Eds. Cham, Switzerland: Springer, 2017, pp. 99–109.

[44] D. Yazdani, J. Branke, M. N. Omidvar, T. T. Nguyen, and X. Yao, "Changing or keeping solutions in dynamic optimization problems with switching costs," in *Proc. Genet. Evol. Comput. Conf.*, 2018, pp. 1095–1102.

[45] H. Fu, B. Sendhoff, K. Tang, and X. Yao, "Finding robust solutions to dynamic optimization problems," in *Proc. Appl. Evol. Comput.*, vol. 7835, 2013, pp. 616–625.

[46] Y. N. Guo, M. Chen, H. Fu, and Y. Liu, "Find robust solutions over time by two-layer multi-objective optimization method," in *Proc. IEEE Congr. Evol. Comput. (CEC)*, 2014, pp. 1528–1535.

[47] Y. Huang, Y. Ding, K. Hao, and Y. Jin, "A multi-objective approach to robust optimization over time considering switching cost," *Inf. Sci.*, vols. 394–395, pp. 183–197, Jul. 2017.

[48] S. Das, S. Maity, B.-Y. Qu, and P. N. Suganthan, "Real-parameter evolutionary multimodal optimization—A survey of the state-of-the-art," *Swarm Evol. Comput.*, vol. 1, no. 2, pp. 71–88, 2011.

[49] R. Cheng, M. Li, K. Li, and X. Yao, "Evolutionary multiobjective optimization-based multimodal optimization: Fitness landscape approximation and peak detection," *IEEE Trans. Evol. Comput.*, vol. 22, no. 5, pp. 692–706, Oct. 2018.

[50] W. Luo, X. Lin, T. Zhu, and P. Xu, "A clonal selection algorithm for dynamic multimodal function optimization," *Swarm Evol. Comput.*, vol. 50, Nov. 2019, Art. no. 100459.

[51] B. Nasiri, M. Meybodi, and M. Ebadzadeh, "History-driven particle swarm optimization in dynamic and uncertain environments," *Neurocomputing*, vol. 172, pp. 356–370, Jul. 2016.

[52] J. J. Grefenstette, "Evolvability in dynamic fitness landscapes: A genetic algorithm approach," in *Proc. Congr. Evol. Comput. (CEC)*, vol. 3, 1999, pp. 2031–2038.

[53] T. M. Blackwell, "Swarms in dynamic environments," in *Proc. Conf. Genet. Evol. Comput.*, vol. 2723, 2003, pp. 19–26.

[54] C. Li, S. Yang, and M. Yang, "An adaptive multi-swarm optimizer for dynamic optimization problems," *Evol. Comput.*, vol. 22, no. 4, pp. 559–594, 2014.

[55] C. Li, M. Mavrovouniotis, S. Yang, and X. Yao, "Benchmark generator for the IEEE WCCI-2014 competition on evolutionary computation for dynamic optimization problems," School Comput. Sci. Informat., De Montfort University, U.K Leicester, U.K., Rep., 2013.

[56] C. Li *et al.*, "Benchmark generator for CEC'2009 competition on dynamic optimization," Center Comput. Intell., Univ. Leicester, Leicester, U.K., Rep., 2008.

[57] T. Zhu, W. Luo, and L. Yue, "Dynamic optimization facilitated by the memory tree," *Soft Comput.*, vol. 19, no. 3, pp. 547–566, 2014.

[58] K. Trojanowski and Z. Michalewicz, "Searching for optima in non-stationary environments," in *Congr. Evol. Comput.*, vol. 3, 1999, pp. 1843–1850.

[59] J. L. Bentley, "Multidimensional binary search trees used for associative searching," *Commun. ACM*, vol. 18, no. 9, pp. 509–517, 1975.

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on October 19,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.

[60] M. N. Omidvar, M. Yang, Y. Mei, X. Li, and X. Yao, "DG2: A faster and more accurate differential grouping for large-scale black-box optimization," *IEEE Trans. Evol. Comput.*, vol. 21, no. 6, pp. 929–942, Dec. 2017.
[61] A. M. Sutton, M. Lunacek, and L. D. Whitley, "Differential evolution and non-separability: Using selective pressure to focus search," in *Proc. 9th Annu. Conf. Genetic. Evol. Comput. (GECCO)*, 2007, pp. 1428–1435.
[62] X. Li, K. Tang, M. N. Omidvar, Z. Yang, and K. Qin, "Benchmark functions for the CEC—2013 special session and competition on large-scale global optimization," Sci. Inf. Technol., RMIT Univ., Melbourne, VIC, USA, Rep., 2013.
[63] R. Tinos and S. Yang, "Continuous dynamic problem generators for evolutionary algorithms," in *Proc. Congr. Evol. Comput.*, 2007, pp. 236–243.
[64] B. Kazimipour, M. N. Omidvar, A. Qin, X. Li, and X. Yao, "Bandit-based cooperative coevolution for tackling contribution imbalance in large-scale optimization problems," *Appl. Soft Comput.*, vol. 76, pp. 265–281, Mar. 2019.



Danial Yazdani (Member, IEEE) received the Ph.D. degree in computer science from Liverpool John Moores University, Liverpool, U.K., in 2018.

He is currently a Postdoctoral Research Fellow with the Department of Computer Science and Engineering, Southern University of Science and Technology, Shenzhen, China. His current research interests include evolutionary algorithms, dynamic optimization problems, and simulation optimization.

Dr. Yazdani was a recipient of the Best Thesis Award from the Faculty of Engineering and Technology, Liverpool John Moores University, and the SUSTech Presidential Outstanding Postdoctoral Award from the Southern University of Science and Technology. He is a member of the IEEE Task Force on Evolutionary Computation in Dynamic and Uncertain Environments.



Mohammad Nabi Omidvar (Member, IEEE) received the Ph.D. degree in computer science from RMIT University, Melbourne, VIC, Australia, in 2016.

He was a Research Fellow with the School of Computer Science, University of Birmingham, Birmingham, U.K. He is currently an Assistant Professor of artificial intelligence in financial services affiliated with Leeds University Business School and School of Computing, University of Leeds, Leeds, U.K. His current research interests include large-scale global optimization, high-dimensional machine learning, and AI for financial services.

Dr. Omidvar is a recipient of the IEEE TRANSACTION ON EVOLUTIONARY COMPUTATION Outstanding Paper Award for his research on large-scale global optimization in 2017, the Australian Postgraduate Award in 2010, and the Best Computer Science Honors Thesis Award from the School of Computer Science and IT, RMIT University. He is the Chair of IEEE Task Force on Large-Scale Global Optimization.



Ran Cheng (Member, IEEE) received the Ph.D. degree from the University of Surrey, Guildford, U.K., in 2016.

He is currently an Assistant Professor with the Department of Computer Science and Engineering, Southern University of Science and Technology, Shenzhen, China.

Dr. Cheng was a recipient of the IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION Outstanding Paper Award in 2018, the IEEE Computational Intelligence Society Outstanding Ph.D. Dissertation Award in 2019, and the IEEE Computational Intelligence Magazine Outstanding Paper Award in 2020. He is an Associate Editor of the IEEE TRANSACTIONS ON ARTIFICIAL INTELLIGENCE. He is the Founding Chair of IEEE Symposium on Model Based Evolutionary Algorithms.



Jürgen Branke (Member, IEEE) received the Ph.D. degree from the University of Karlsruhe, Karlsruhe, Germany, in 2000.

He is a Professor of operational research and systems with the Warwick Business School, University of Warwick, Coventry, U.K. He has been an Active Researcher in the area of evolutionary optimization since 1994 and has published more than 190 papers in international peer-reviewed journals and conferences. His research interests include multiobjective optimization, handling of uncertainty in optimization, dynamically changing optimization problems, simulation-based optimization, and the design of complex systems.

Prof. Branke is an Editor of the ACM TRANSACTIONS ON EVOLUTIONARY LEARNING AND OPTIMIZATION, an Area Editor of the *Journal of Heuristics* and the *Journal on Multicriteria Decision Analysis*, and an Associate Editor of the IEEE TRANSACTIONS IN EVOLUTIONARY COMPUTATION and *Evolutionary Computation Journal*.



Trung Thanh Nguyen received the Ph.D. degree from the University of Birmingham, Birmingham, U.K., in 2011.

He has been a Reader of operational research with Liverpool John Moores University, Liverpool, U.K., since 2015, where he has been a Co-Director of the Liverpool Logistics, Offshore, and Marine Research Institute since 2019. He has led more than 20 research projects in transport/logistics. He has published about 50 peer-reviewed articles. He co-organized six leading conferences, was a TPC member of more than 30 international conferences, edited eight books, and gave speeches to many conferences/events. His current research interests include operational research/dynamic optimization with a particular application to logistics/transport problems.



Xin Yao (Fellow, IEEE) received the Ph.D. degree from the University of Science and Technology of China, Hefei, China, in 1990.

He is the Chair Professor of computer science with the Southern University of Science and Technology, Shenzhen, China, and a part-time Professor of computer science with the University of Birmingham, Birmingham, U.K. His major research interests include evolutionary computation and ensemble learning and their applications to software engineering.

Dr. Yao received the Prestigious Royal Society Wolfson Research Merit Award in 2012, the IEEE CIS Evolutionary Computation Pioneer Award in 2013, and the IEEE Frank Rosenblatt Award in 2020. His paper on evolving artificial neural networks won the 2001 IEEE Donald G. Fink Prize Paper Award. He also won the IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION Outstanding Paper Awards in 2010, 2016, and 2017, the IEEE TRANSACTIONS ON NEURAL NETWORKS Outstanding Paper Award in 2011, and many other best paper awards. He was a Distinguished Lecturer of the IEEE Computational Intelligence Society. He was a President of IEEE CIS from 2014 to 2015 and the Editor-in-Chief of the IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION from 2003 to 2008.

Authorized licensed use limited to: Institut Teknologi Sepuluh Nopember. Downloaded on October 19,2026 at 01:56:39 UTC from IEEE Xplore. Restrictions apply.